

BOGAPURAPU SAI SRINIVAS

Tuni, Andhra Pradesh

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SUMMARY

Systems enthusiast with strong proficiency in C, C++, and Python, and experience in low-level programming, interrupt handling, and embedded debugging. Knowledgeable in 8086 microprocessor and 8051 microcontroller architectures. Skilled in PCB design (Altium), simulation (Proteus), and optimizing systems for real-time and memory constraints, with hands-on experience in STM32 bare-metal programming on ARM Cortex-M.

EDUCATION

Degree	Specialization	Institute	Year	CGPA
B.Tech	ECE	RGUKT Nuzvid	2022–2026t	7.49
PUC	Physics, Chemistry, & Mathematics	RGUKT Nuzvid	2020–2022	9.47
10th Class	General Curriculum (SSC)	Sri Chaitanya	2020	10.0

PROJECTS

- **RFID-Based IoT Cloud Automation** (IoT, RFID, Cloud Integration)
 - Designed and implemented an RFID-based system for automated access control and monitoring.
 - Integrated IoT devices with a cloud platform to enable real-time data logging, tracking, and analysis.
 - Developed firmware for RFID module interfacing and seamless data transmission to the cloud.
- **Distance-Controlled Mobile Robot** (Embedded Systems)
 - Designed and developed a mobile robot capable of traveling a user-defined distance based on mobile app input.
- **Student Marks Prediction System** (Python, Machine Learning)
 - Developed a predictive model to estimate student performance based on historical academic and demographic data.
 - Implemented and compared multiple ML algorithms (e.g., Linear Regression, Random Forest) to select the best-performing model.
 - Enhanced accuracy through hyperparameter tuning and cross-validation techniques.
- **5G Channel Estimation using Machine Learning** (5G NR, Python, ML)
 - Developed a machine learning-based model for accurate channel estimation in 5G NR systems.
 - Simulated wireless channels (AWGN, Rayleigh fading) and generated training datasets.
 - Compared performance with traditional estimation methods (LS, MMSE) using **BER and MSE metrics**.
 - Improved estimation accuracy under noisy conditions using data-driven approaches.
 - Implemented modulation schemes (QPSK/QAM) and OFDM-based transmission for realistic simulation.
- **Autonomous Path Planning & Navigation using Remote-Sensed Imagery** (Computer Vision, Robotics)
 - Developed an autonomous path navigation system using remote-sensed imagery integrating computer vision and robotics.
 - Implemented image processing pipeline using OpenCV for obstacle detection, arena extraction, and path identification.
 - Designed and applied **camera calibration algorithms** to correct lens distortion and improve spatial accuracy.
 - Implemented **global path planning algorithms** (A*, Dijkstra) for optimal route computation.

VOLUNTARY WORK EXPERIENCE

- **Teckzite 2k25 – Robotics Hackathon Coordinator**
 - Coordinated and managed the Robotics Hackathon event for Teckzite 2k25, overseeing planning, scheduling, and execution.
 - Collaborated with organizing teams, mentors, and participants to deliver a successful and engaging robotics competition.
- **Finsara 2k25 – Overall Coordinator (ECE Farewell Event)** *March 2025 – Apr 2025*
 - Served as the overall coordinator for Finsara 2k25, the farewell event for the ECE department.
 - Managed end-to-end event planning, logistics, and execution, ensuring a smooth and memorable experience for attendees.
- **Student Mess Committee Head** *Sep 2024 – Apr 2025*
 - Led the student mess committee, overseeing daily operations and quality control for meal services.
 - Coordinated with vendors, staff, and administration to ensure timely and hygienic food delivery.

TECHNICAL SKILLS

- **Programming:** Proficient in C, C++, and Python; experience in Machine Learning model development and embedded programming for microcontrollers (Arduino ATmega328, ESP8266, ESP32).
- **Embedded PCB Designing:** Experienced in PCB design using Altium, including schematic design, component placement, and embedded hardware integration.
- **Embedded Systems & Microcontrollers:** experience with **STM32 (Cortex-M4)** using **bare-metal programming** (no HAL). Deep understanding of ARM Cortex-M architecture including memory map, register-level access, GPIO, RCC, NVIC, SysTick, and interrupts. Experience with Arduino (ATmega328), ESP8266, ESP32, and 8086 microprocessor programming (Assembly).
- **5G Physical Layer Design:** Explored concepts such as OFDM, CP-OFDM, DFT-s-OFDM, channel coding, modulation schemes (QPSK, QAM), and MIMO.
- **Communication Systems:** Solid foundation in analog and digital communication — AM, FM, PM, pulse modulation, sampling, quantization, PCM, line coding, digital modulation (ASK, PSK, FSK, QAM), and error control coding — applied to understanding advanced wireless systems (4G/5G).

CERTIFICATIONS

- **MicroProcessor and Interfacing** – IIT Guwahati *Apr 2025*
- **Introduction to Wireless and Cellular Communication** – IIT Madras *Sep 2025*
- **Sensors and Actuators** – IISC *May 26*
- **Electronic Systems Design: Hands-on Circuits and PCB Design with CAD Software** – IIT Madras *Sep 25*
- **Strategies for Sustainable Design** – IIT Hyderabad *April 2025*
- **Embedded Systems** – SkillDzire *Jun 2025*
- **Timing Advance in 5G** – Udemy *Jun 2025*
- **Circuits and Electronics** – edX *Mar 2024*
- **Machine Learning with Python** – Coursera *Aug 2023*
- **AI for Everyone: Master the Basics** – edX

ACHIEVEMENTS

- Selected among the **Top 100 candidates** for the *100 Days 5G Physical Layer Design* course by Sakshama Ghoslya.
- **Winner** – Robo Soccer Event, Teckzite 2k24.
- Qualified GATE 2026 EC Paper
- Secured **1st place** in the *Ramanujan Maths Olympiad* (Mandal level).